KCET-2017 (Physics)

1. A substance of mass 49.53 g occupies 1.5 cm^3 of volume. The density of the substance (in g cm^3) with correct number of significant figures is
   a. 3.302
   b. 3.300
   c. 3.3
   d. 3.30

2. A car moving with a velocity of 20 ms^{-1} is stopped in a distance of 40 m. If the same car is travelling at double the velocity, the distance travelled by it for same retardation is
   a. 640 m
   b. 320 m
   c. 1280 m
   d. 160 m

3. The angle between velocity and acceleration of a particle describing uniform circular motion is
   a. 45°
   b. 60°
   c. 90°
   d. 180°

4. If $\vec{A} = 2\hat{i} + 3\hat{j} + 8\hat{k}$ is perpendicular to $\vec{B} = 4\hat{j} - 4\hat{i} + \alpha\hat{k}$, then the value of '$\alpha$' is
   a. $\frac{1}{2}$
   b. $-\frac{1}{2}$
   c. 1
   d. –1

5. A body of mass 50 kg, is suspended using a spring balance inside a lift at rest. If the lift starts falling freely, the reading of the spring balance is
   a. = 50 kg
   b. > 50 kg
   c. < 50 kg
   d. =0

6. A motor pump lifts 6 tones of water from a well of depth 25m to the first floor of height 35 m from the ground floor in 20 minutes. The power of the pump (in kW) is [g = 10 ms^{-2}]
   a. 3
   b. 6
   c. 1.5
   d. 12

7. Two balls are thrown simultaneously in air. The acceleration of the centre of mass of the two balls when in air
   a. depends on the masses of the two balls
   b. depends on the speeds of the two balls
   c. is equal to g (Acceleration due to gravity)
   d. depends on the direction of motion of the two balls

8. The value of acceleration due to gravity at a depth of 1600 km is equal to [Radius of earth = 6400 km]
   a. 9.8 ms^{-2}
   b. 19.6 ms^{-2}
   c. 4.9 ms^{-2}
   c. 7.35ms^{-2}

Copyright © Think and Learn Pvt. Ltd.

9. 'Young's modulus is defined as the ratio of
   a. tensile stress and longitudinal strain
   b. hydraulic stress and hydraulic strain
   c. shearing stress and shearing strain
   d. bulk stress and longitudinal strain

10. 'Hydraulic lift' works on the basis of
   a. Stoke's law
   b. Toricelli's law
   c. Pascal's law
   d. Bernoulli's law

11. The S.I. unit of specific heat capacity is
   a. $J\ mol^{-1}\ K^{-1}$
   b. $J\ kg^{-1}\ K^{-1}$
   c. $J\ K^{-1}$
   d. J kg

12. For which combination of working temperatures, the efficiency of 'Carnot's engine' is the least?
   a. 60K, 40 K
   b. 40K, 20K
   c. 80K, 60K
   d. 100K, 80K

13. The mean energy of a molecule of an ideal gas is
   a. 2 KT
   b. $\frac{3}{2}$ KT
   c. KT
   d. $\frac{1}{2}$ KT

14. Two simple pendulums A and B are made to oscillate simultaneously and it is found that A completes 10 oscillations in 20 sec and B completes 8 oscillations in 10 sec. The ratio of the lengths of A and B is
   a. $\frac{8}{5}$
   b. $\frac{64}{25}$
   c. $\frac{5}{4}$
   d. $\frac{25}{64}$

15. The waves set up in a closed pipe are
   a. Transverse and progressive
   b. Longitudinal and stationary
   c. Transverse and stationary
   d. Longitudinal and progressive

16. Two spheres of electric charges +2 nC and −8 nC are placed at a distance'd' apart. If they are allowed to touch each other, what is the new distance between them to get a repulsive force of same magnitude as before?
   a. $\frac{4d}{3}$
   b. $\frac{3d}{4}$
   c. d
   d. $\frac{d}{2}$

Copyright © Think and Learn Pvt. Ltd.

17. Three point charges of +2q, + 2q and –4q are placed at the corner A, B and C of an equilateral triangle ABC of side 'x'. The magnitude of the electric dipole moment of this system is
    a. 2 qx
    b. $2\sqrt{3}$ qx
    c. $3\sqrt{2}$ qx
    d. 3 qx

18. 4×10^{10} electrons are removed from a neutral metal sphere of diameter 20 cm placed in air. The magnitude of the electric field (in NC^{-1}) at a distance of 20 cm from its centre is
    a. 5760
    b. 1440
    c. 640
    d. zero

19. Two point charges A =+3 nC and B = 1nC are placed 5 cm apart in air. The work done to move charge B towards A by 1 cm is
    a. 1.35×10^{-7}J
    b. 2.7×10^{-7} J
    c. 2.0×10^{-7} J
    d. 12.1×10^{-7}J

20. A system of 2 capacitor of capacitance 2μF and 4μF is connected in series across a potential difference of 6V. The electric charge and energy stored in the system are
    a. 10μ C and 30 μJ
    b. 36 μC and 108 μJ
    c. 8 μC and 24 μJ
    d. 1μC and 2 μJ

21. The minimum value of effective capacitance that can be obtained by combining 3 capacitors of capacitances 1pF, 2pF and 4pF is
    a. $\frac{4}{7}$pF
    b. 1pF
    c. $\frac{7}{4}$pF
    d. 2pF

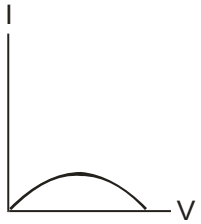
22. A cylindrical conductor of diameter 0.1 mm carries a current of 90 mA. The current density (in Am^{-2}) is ($\pi \simeq 3$)
    a. 1.2×10^{7}
    b. 2.4×10^{7}
    c. 3×10^{6}
    d. 6×10^{6}

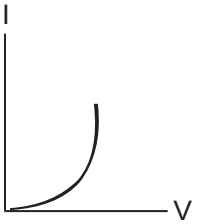
23. A piece of copper is to be shaped into a conducting wire of maximum resistance. The suitable length and diameter are _____ and _____respectively.
    a. L and d
    b. 2 L and d
    c. L/2 and 2 d
    d. L and d/2

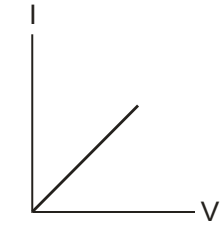
Copyright © Think and Learn Pvt. Ltd.

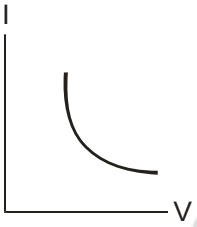
KCET-2017 (Physics)

24. Of the following graphs, the one that correctly represents the I –V characteristics of a 'Ohmic device' is

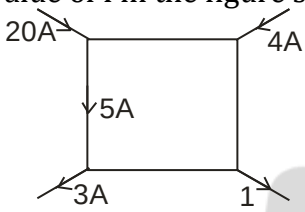
a. 

b. 

c. 

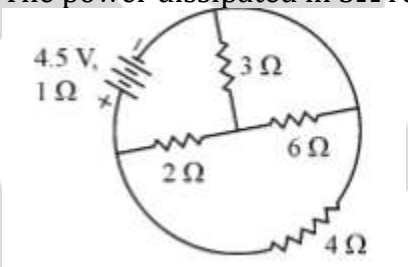
d. 

25. The value of I in the figure shown below is



a. 8A
b. 21A
c. 19A
d. 4A

26. The power dissipated in 3Ω resistance in the following circuit is



a. – 261 kJ
b. + 103 kJ
c. +261 kJ
d. –103 kJ

27. In metre bridge experiment, with a standard resistance in the right gap and a resistance coil dipped in water (in a beaker) in the left gap, the balancing length obtained is 'I'. If the temperature of water is increased, the new balancing length is

a. > 1
b. < 1
c. = 1
d. = 0

28. A proton, a deuteron and an α - particle are projected perpendicular to the direction of a uniform magnetic field with same kinetic energy. The ratio of the radii of the circular paths described by them is

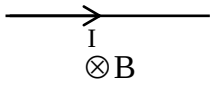
a. $1:\sqrt{2}:1$
b. $1:\sqrt{2}:\sqrt{2}$
c. $\sqrt{2}:1:1$
d. $\sqrt{2}:\sqrt{2}:1$

Copyright © Think and Learn Pvt. Ltd.

29. A galvanometer of resistance 50 Ω is connected to a battery of 3V along with a resistance of 2950 Ω in series shows full-scale deflection of 30 divisions. The additional series resistance required to reduce the deflection to 20 divisions is
    a. 1500 Ω
    b. 4440 Ω
    c. 7400Ω
    d. 2950 Ω

30. The magnetic field at the center of a current carrying loop of radius 0.1 m is $5\sqrt{5}$ times that at a point along its axis. The distance of this point from the centre of the loop is
    a. 0.2 m
    b. 0.1m
    c. 0.05m
    d. 0.25m

31. A straight wire of length 50 cm carrying a current of 2.5 A is suspended in mid-air by a uniform magnetic field of 0.5 T (as shown in figure). The mass of the wire is (g = 10 ms^{-2})



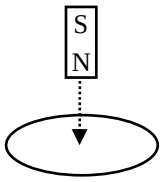
    a. 62.5 gm
    b. 250 gm
    c. 125 gm
    d. $Cu_2O + FeS$

32. Which of the following properties is 'False' for a bar magnet?
    a. Its poles cannot be separated.
    b. It points in North-South direction when suspended
    c. It's like poles repel and unlike poles attract.
    d. It doesn't produce magnetic field

33. A magnetic dipole of magnetic moment 6×10^{-2} $Am^2$and moment of inertia 12×10^{-6} kgm^2 performs oscillation in a magnetic field of 2×10^{-2} T. The time taken by the dipole to complete 20 oscillations is
    ($\pi \simeq 3$)
    a. 36 s
    b. 6s
    c. 12 s
    d. 18 s

34. The susceptibility of a ferromagnetic substance
    a. >>1
    b. > 1
    c. < 1
    d. zero

35. A bar magnet is allowed to fall vertically through a copper coil placed in a horizontal plane. The magnet falls with a net acceleration



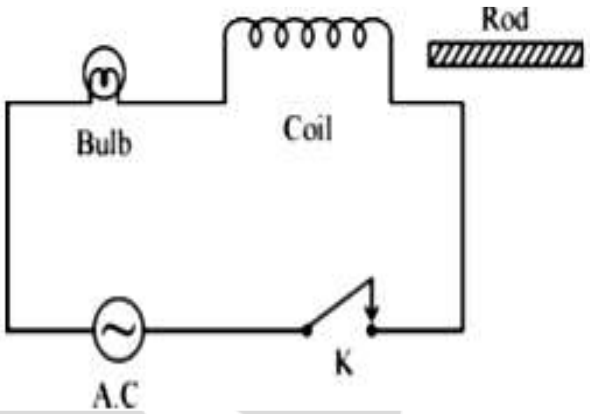
    a. = g
    b. > g
    c. < g
    d. zero

Copyright © Think and Learn Pvt. Ltd.

36. The working of magnetic braking of trains is based on
    a. Alternating current
    b. Eddy current
    c. Steady current
    d. Pulsating current

37. A jet plane of wing span 20 m is travelling towards west at a speed of 400 ms^{-1}. If the earth's total magnetic field is 4×10^{-4} T and the dip angle is 30°, at that place, the voltage difference developed across the ends of the wing is
    a. 1.6 V
    b. 3.2 V
    c. 0.8 V
    d. 6.4 V

38. In the A.C. circuit shown, keeping 'K' pressed, if an iron rod is inserted into the coil, the bulb in the circuit,



    a. Glows more brightly
    b. Glows less brightly
    c. Glows with same brightness (as before the rod is inserted)
    d. Gets damaged

39. The output of a step down transformer is measured to be 48 V when connected to a 12 w bulb. The value of peak current is
    a. $\frac{1}{\sqrt{2}}$A
    b. $\sqrt{2}$A
    c. $\frac{1}{2\sqrt{2}}$A
    d. $\frac{1}{4}$A

40. A coil of inductive reactance $1/\sqrt{3}\Omega$ and resistance 1 Ω is connected to a 200 V, 50 Hz A.C. supply. The time lag between maximum voltage and current is
    a. $\frac{1}{300}$s
    b. $\frac{1}{600}$s
    c. $\frac{1}{500}$s
    d. $\frac{1}{200}$s

Copyright © Think and Learn Pvt. Ltd.

KCET-2017 (Physics)

41. If $\vec{E}$ and $\vec{B}$ represent electric and magnetic field vectors of an electromagnetic wave, the direction of propagation of the wave is along
    a. $\vec{E}$
    b. $\vec{B}$
    c. $\vec{E} \times \vec{B}$
    d. $\vec{B} \times \vec{E}$

42. According to Cartesian sign convention, in ray optics
    a. all distances are taken positive
    b. all distances are taken negative
    c. all distances in the direction of incident ray are taken positive
    d. all distances in the direction of incident ray are taken negative

43. A linear object of height 10 cm is kept in front of a concave mirror of radius of curvature 15 cm, at a distance of 10 cm. The image formed is
    a. magnified
    b. magnified and inverted
    c. diminished and erect
    d. diminished and inverted

44. During scattering of light, the amount of scattering is inversely proportional to _____ of wavelength of light,
    a. cube
    b. square
    c. fourth power
    d. half

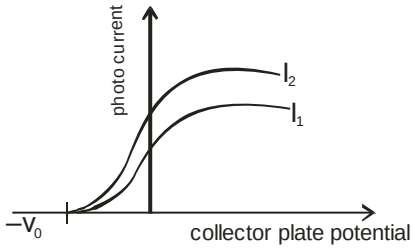
45. In Young's double-slit experiment if yellow light is replaced by blue light, the interference fringes become
    a. wider
    b. narrower
    c. brighter
    d. darker

46. According to Huygens' principle, during refraction of light from air to a denser medium
    a. Wavelength and speed decrease
    b. Wavelength and speed increase
    c. Wavelength increases but speed decreases
    d. Wavelength decreases but speed increases

47. In a system of two crossed polarisers, it is found that the intensity of light from the second polariser is half from that of first polariser. The angle between their pass axes is
    a. 45°
    b. 60°
    c. 30°
    d. 0°

Copyright © Think and Learn Pvt. Ltd.

48. From the following graph of photo current against collector plate potential, for two different intensities of light I_1 and I_2 , one can conclude



a. $I_1 = I_2$
b. $I_1 > I_2$
c. $I_1 < I_2$
d. Comparison is not possible

49. A particle is dropped from a height 'H'. The de'Broglie wavelength of the particle depends on height as
a. H
b. H^0
c. $H^{1/2}$
d. $H^{-1/2}$

50. The scientist who is credited with the discovery of 'nucleus' in an atom is
a. J.J. Thomson
b. Rutherford
c. Niels Bohr
d. Balmer

51. The energy (in eV) required to excite an electron from n = 2 to n = 4 state in hydrogen atom is
a. + 2.55
b. – 3.4
c. – 0.85
d. + 4.25

52. In a nuclear reactor the function of the Moderator is to decrease
a. Number of neutrons
b. Speed neutrons
c. Escape of neutrons
d. Temperature of the reactor

53. The particles emitted in the decay of $^{238}_{92}U$ to $^{234}_{92}U$
a. 1 α and 2 β
b. 1 α only
c. 1 α and 1β
d. 2 α and 2 β

54. The mass defect of $^{4}_{2}He$ is 0.03 μ. The binding energy per nucleon of helium (in MeV) is
a. 27.93
b. 6.9825
c. 2.793
d. 69.825

55. The energy gap in case of which of the following is less than 3 eV?
a. copper
b. Iron
c. Zener diode
d. Germanium

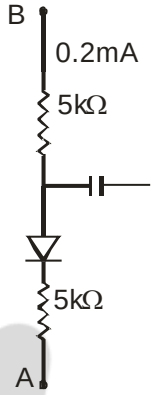
Copyright © Think and Learn Pvt. Ltd.

KCET-2017 (Physics)

56. Which of the following semi-conducting devices is used as a voltage regulator?
    a. Photo diode
    b. LASER diode
    c. Zener diode
    d. Solar cell

57. In the three parts of a transistor, 'Emitter is of
    a. Moderate size and heavily doped
    b. Large size and lightly doped
    c. Thin size and heavily doped
    d. Large size and moderately doped

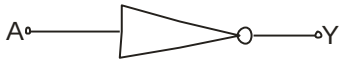
58. In the figure shown, if the diode forward voltage drop is 0.2 V, the voltage difference between A and B is



    a. 1.3 V
    b. 2.2 V
    c. 0
    d. 0.5 V

59. Which of the following logic gates is considered as 'universal'?

    a. 

    b. 

    c. 

    d. 

60. A basic communication system consists of
    (a) Transmitter
    (b) Information source
    (c) User of information
    (d) Channel
    (e) Receive
    a. a, b, c, d and e
    b. b, a, d, e and c
    c. b, d, a, c and e
    d. b, e, a, d and c

Copyright © Think and Learn Pvt. Ltd.

KCET-2017 (Physics)

ANSWER KEYS

1. (G)	2. (d)	3. (c)	4. (b)	5. (d)	6. (a)	7. (c)	8. (d)	9. (a)	10. (c)
11. (b)	12. (d)	13. (b)	14. (b)	15. (b)	16. (b)	17. (b)	18. (b)	19. (a)	20. (c)
21. (a)	22. (a)	23. (d)	24. (c)	25. (b)	26. (a)	27. (a)	28. (a)	29. (b)	30. (a)
31. (a)	32. (d)	33. (c)	34. (a)	35. (c)	36. (b)	37. (a)	38. (b)	39. (c)	40. (b)
41. (c)	42. (c)	43. (b)	44. (c)	45. (b)	46. (a)	47. (a)	48. (c)	49. (d)	50. (b)
51. (a)	52. (b)	53. (a)	54. (b)	55. (d)	56. (c)	57. (a)	58. (b)	59. (d)	60. (b)

* G – Indicates One GRACE MARK awarded for the question number.

BYJU'S

Copyright © Think and Learn Pvt. Ltd.

KCET-2017 (Physics)

Solution

1. (G)
   Bonus

2. (d)
   Let; initial speed of car $\Rightarrow$ u = 20 m/s
   Final speed of car; V = 0 m/s
   Retardation = a
   Case –I $\Rightarrow V^2 = u^2 + 2as$
   $\Rightarrow 0 = u^2 + 2as$
   $\Rightarrow \frac{-400}{2\times 40} = a$
   $\Rightarrow [a = -5\ m/s^2]$
   Case – II u = 40 m/s
   v = 0 m/s
   $a = -5\ m/s^2$
   s =?
   $\Rightarrow V^2 = u^2 + 2as$
   $\Rightarrow 0 = (40)^2 + 2(-5).S$
   $\Rightarrow S = \frac{1600}{10}$
   $\Rightarrow S = 160\ m$

3. (c)
   In circular motion, velocity is always tangential to the circular path. Also in uniform circular motion, tangential acceleration is zero. Hence, the net acceleration will be centripetal acceleration which always acts towards the centre of the circular path.
   Hence, velocity and acceleration of a particle describing uniform circular motion are at right angle to each other.

4. (b)
   When vectors are perpendicular to each other, then their dot product is zero.
   $\Rightarrow \vec{A}.\vec{B} = 0$
   $\Rightarrow \left(2\hat{i} + 3\hat{j} + 8\hat{k}\right).\left(-4\hat{i} + 4\hat{j} + \alpha\hat{k}\right) = 0$
   $\Rightarrow -8 + 12 + 8\alpha = 0$
   $\Rightarrow 8\alpha = -4$
   $\alpha = \frac{-1}{2}$
   $\therefore$ (B)

5. (d)
   As the lift is falling freely, therefore its acceleration = g (downwards)
   Let, the force acting on body due to spring is "F"
   $\therefore$ (50g) – F = (50) a
   $\Rightarrow$ 50g – F = 50 g
   $\Rightarrow$ F = 0
   Thus, no force is acting on the body due to spring. So, the reading of spring balance is zero.

Copyright © Think and Learn Pvt. Ltd.

6. (a)

$$\text{Power} = \frac{\text{work done}}{\text{time taken}}$$

$$P = \frac{mgh}{t}$$

$$P = \left(\frac{m}{t}\right)gh$$

$h = 25 + 35 = 60\text{m}$.

$$\text{Flow rate} = \left(\frac{m}{t}\right) = 6 \text{ tones / } 20 \text{ min}$$

$$= \frac{6000}{20\times 60}\frac{\text{kg}}{\text{sec}} = 5\text{kg/sec}$$

$\therefore$ $P = 5 \times 10 \times 60$

$= 3000$ w

$= 3$ kw

7. (c)

As there is no force in horizontal direction, so there will be no acceleration of COM in horizontal direction.

As for vertical direction only gravitational force acts on both the balls, hence acceleration of COM will be equal to acceleration due to gravity which is 'g'.

8. (d)

Acceleration due to gravity at a depth "d":

$$g_d = g\left(1 - \frac{d}{R_e}\right)$$

$d \rightarrow 1600\text{km}$

$R_e \rightarrow 6400$ km

$g \rightarrow 9.8$ m/s^2

$$\Rightarrow g_d = 9.8\left(1 - \frac{1600}{6400}\right)$$

$\Rightarrow g_d = 7.35$ m/s^2

9. (a)

Young's modulus is defined as the ratio of tensile stress to tensile strain or longitudinal strain

10. (c)

Hydraulic lift works on the principle of equal transmission throughout a fluid i.e. pascal's law.

11. (b)

Copyright © Think and Learn Pvt. Ltd.

Specific heat capacity: $S = \left(\frac{Q}{m\Delta T}\right)$

$Q \rightarrow$ Amount of heat absorbed
$M \rightarrow$ Mass of the substance
$\Delta T \rightarrow$ Change in temperature of the substance
$\therefore$ SI unit of $S \rightarrow J\ Kg^{-1}\ K^{-1}$

12. (d)

Efficiency of carnot's engine; $\eta = 1 - \frac{T_{Lower}}{T_{Higher}}$

Now, let's check options.

$\therefore$ (A) $\rightarrow \eta_A = 1 - \frac{40}{60} = 0.33$

(B) $\rightarrow \eta_B = 1 - \frac{20}{40} = 0.5$

(C) $\rightarrow \eta_c = 1 - \frac{60}{80} = 0.25$

(D) $\rightarrow \eta_d\ 1 - \frac{80}{100} = 0.2$

13. (b)

Mean energy of a molecule of an ideal gas:

$E = \frac{1}{2}mV_{rms}^2$

Rms velocity of molecules: $V_{rms} = \sqrt{\frac{3KT}{m}}$

$\therefore$ $E = \frac{1}{2} \times m \times \left(\frac{3KT}{m}\right)$

$\therefore$ $E = \frac{3}{2}KT$

14. (b)

Time period of simple pendulum: $T = 2\pi\sqrt{\frac{\ell}{g}}$

$\Rightarrow$ $\frac{\ell_A}{\ell_B} = \left[\frac{T_A}{T_B}\right]^2$

Time period of A: $T_A = \frac{20}{10} = 2$ sec

Time period of B: $T_B = \frac{10}{8} = \frac{5}{4}$ sec

$\therefore$ $\frac{\ell_A}{\ell_B} = \frac{64}{25}$

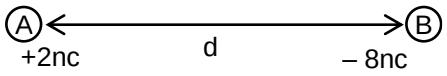
Copyright © Think and Learn Pvt. Ltd.

15. (b)

The waves set up in a closed pipe are longitudinal and stationary because of superposition of incident wave and reflected wave.

16. (b)

Case – I:



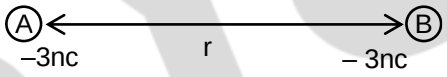
$$F = \frac{Kq_Aq_B}{r^2}$$

$$\Rightarrow \quad F = \frac{K.2.8}{d^2}$$

$$\Rightarrow \quad F = \frac{16K}{d^2}$$

Case – II: (A)(B)

$$q_A = q_B = \frac{2-8}{2} = \frac{-6}{2} = -3nc$$



$$F' = \frac{K(3)(3)}{r^2}$$

But $$F = \frac{16k}{d^2}$$

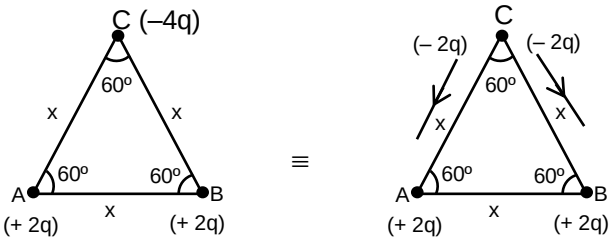
$$\Rightarrow \quad \frac{16k}{d^2} = \frac{9k}{r^2}$$

$$\Rightarrow \quad r^2 = \frac{9}{16}d^2$$

$$\Rightarrow \quad r = \frac{3}{4}d$$

Copyright © Think and Learn Pvt. Ltd.

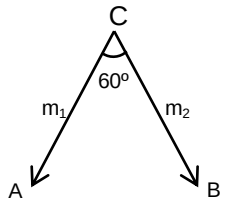
17. (b)



As evident from the diagram above, we can consider two dipoles one CA and CB.

M_1: Dipole moment of CA = 2q x (from C to A)

M_2: Dipole moment of CB = 2qx (from C to B)



$\therefore$ Net dipole moment $= \sqrt{m_1^2 + m_2^2 + 2m_1m_2\cos 60^\circ}$

$= \sqrt{m_1^2 + m_2^2 + 2m_1m_2 \times \frac{1}{2}}$

$= \sqrt{m_1^2 + m_2^2 + m_1m_2}$

$= \sqrt{4q^2x^2 + 4q^2x^2 + 4q^2x^2}$

$= 2\sqrt{3}qx$

18. (b)

Total charge removed from sphere = ne

$= (4 \times 10^{10}) \times (1.6 \times 10^{-19})$

$= 6.4 \times 10^{-9}$ C

Distance of point from centre of sphere = 20 cm =0.2 m

$\therefore$ Electric field: $E = \frac{kq}{r^2}$

$\Rightarrow$ $E = \frac{9\times10^9 \times (6.4\times10^{-9})}{(0.2)^2}$

$= 1440$ N/C

Copyright © Think and Learn Pvt. Ltd.

19. (a)

Potential energy of system: $u = \frac{KQ_AQ_B}{d}$

Initial distance between charges $\rightarrow$ 5 cm = 0.05m

$$U_{initial} = \frac{(9\times10^{9})\times(3\times10^{-9})\times(1\times10^{-9})}{0.05}$$

$= 5.4 \times 10^{-7}$ J

Final distance between charged $\rightarrow$ 5 – 1 = 4 cm =0.04m

$$\therefore \quad U_{final} = \frac{(9\times10^{9})\times(3\times10^{-9})\times(1\times10^{-9})}{0.04}$$

$= 6.75 \times 10^{-7}$ J

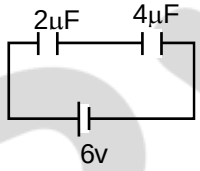
Work done: $W = u_f - u_i$

$\therefore \quad W = (6.75 - 5.4) \times 10^{7}$

$= 1.35 \times 10^{7}$ J

20. (c)

Equivalent capacitance: $C_{eq} = \frac{C_1C_2}{C_1+C_2} = \frac{2\times4}{2+4}$



$C_{eq} = \frac{4\mu F}{3}$

$\therefore$ Charge: $q = cv$

$\Rightarrow \quad Q = \frac{4}{3} \times 6 = 8\ \mu C$

Energy stored: $E = \frac{1}{2}CV^2 = \frac{1}{2} \times \frac{4}{3}(6^2) = 24\ \mu J$

21. (a)

Minimum capacitance can be obtained by connecting these three capacitors in series

$$\Rightarrow \quad \frac{1}{C_{eq}} = \frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}$$

$$= \frac{1}{1}+\frac{1}{2}+\frac{1}{4}$$

$$= \frac{7}{4}$$

$$\Rightarrow \quad C_{eq} = \frac{4}{7}\ pF$$

Copyright © Think and Learn Pvt. Ltd.

22. (a)

Current density: $J = \frac{I}{A}$

Current: $I \longrightarrow 90\ mA = 90 \times 10^{-3} A$
Diameter of wire: $d = 0.1mm = 10^{-4}\ m^2$

Cross section area of wire; $A = \frac{\pi d^2}{4}$

$$A = \frac{3\times(10^{-4})^2}{4} = 0.75 \times 10^{-8}\ m^2$$

$$\therefore \quad J = \frac{90\times10^{-3}}{0.75\times10^{-8}} = 1.2 \times 10^{7}\ A/m^2$$

23. (d)

Resistance of copper wire: $R = \frac{\rho L}{A}$

$\rho \rightarrow$ Resistivity of material and is constant
$L \rightarrow$ Length of wire
$A \rightarrow$ Cross sectional area of conductor

$$A = \frac{\pi d^2}{4}$$

$D \rightarrow$ diameter of wire

$$\Rightarrow \quad R \propto \frac{L}{d^2}$$

Now, let's check the options

(A) $R_A \propto \frac{L}{d^2}$

(B) $R_B \propto 2\frac{L}{d^2}$

(C) $R_B \propto \frac{1}{8}\frac{L}{d^2}$

(D) $R_D \propto 8\ \frac{L}{d^2}$

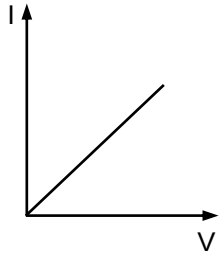
24. (c)

Ohmic device, obey ohm's law, i.e.

$V = IR$

$\Rightarrow \quad V \propto I \quad (R \rightarrow \text{constant})$

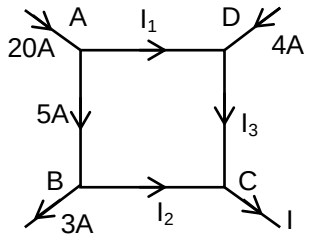
$\therefore$ I – V characteristic will be straight line



Copyright © Think and Learn Pvt. Ltd.

25. (b)

We will use concepts of KCL i.e. $I_{in} = I_{out}$ at a junction



Using KCL at A

$20 = 5 + I_1$

$\Rightarrow$ $I_1 = 15A$

Using KCL at B $5 = 3 + I_2$

$\Rightarrow$ $I_2 = 2A$

Using KCL at D $I_1 + 4 = I_3$

$I_3 = 15 + 4$

$\Rightarrow$ $I_3 = 19A$

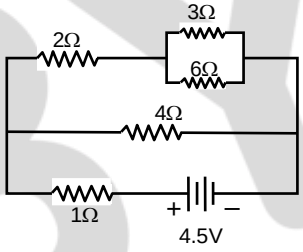
Using KCL at C $I = I_2 + I_3$

$\Rightarrow$ $I = 2 + 19$

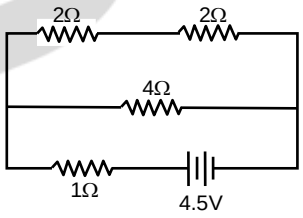
$\Rightarrow$ $I = 21A$

26. (a)

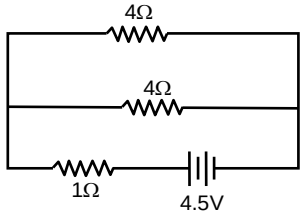
The given circuit can be redrawn as



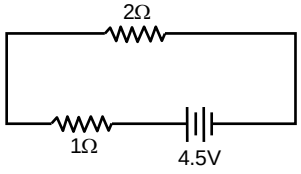
$\Rightarrow$



$\Rightarrow$



$\Rightarrow$

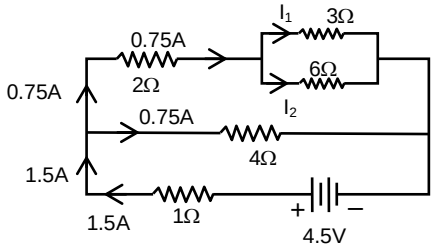


$\Rightarrow$ $I = \frac{V}{R} = \frac{4.5}{2+1}$

$\Rightarrow$ $I = \frac{4.5}{3} = 1.5A$

Copyright © Think and Learn Pvt. Ltd.

Again



$$I_1 = I \times \frac{6}{3+6} = 0.75 \times \frac{6}{9}$$

$\Rightarrow \quad I_1 = 0.5A$

Power in $3\Omega = I_1^2 R = (0.5)^2 \times 3 = 0.75W$

27. (a)

$$\frac{R_{unknown}}{R_{standard}} = \frac{\ell}{(1-\ell)}$$

As temperature increases Þ Resistance increases

$\Rightarrow \quad \frac{R_{unknown}}{R_{standard}}$ Increases $\quad \Rightarrow \quad \frac{\ell}{(1-\ell)}$ Increases

$\Rightarrow \quad (1-\ell)$ should decrease $\quad \Rightarrow \quad (\ell > 1)$

28. (a)

$$v = \frac{qBr}{m} \Rightarrow r = \frac{mV}{qB}$$

$P = mV = \sqrt{2mE}$, E is same (kinetic energy)

$$r = \frac{\sqrt{2mE}}{qB}$$

$$r \propto \frac{\sqrt{m}}{q}$$

Or $r = \sqrt{\frac{m}{q^2}}$

$$r_p : r_d : r_\alpha = \sqrt{\frac{m}{e^2}} : \sqrt{\frac{2m}{e^2}} : \sqrt{\frac{4m}{4e^2}}$$

$$= 1 : \sqrt{2} : 1$$

Copyright © Think and Learn Pvt. Ltd.

29. (b)

Total internal resistance = G + R

= 50 + 2950 = 3000Ω

Current; $I = \frac{3}{3000} = 10^{-3}A = 1$ mA.

If the deflection is to be reduced to 20 divisions

$\Rightarrow \quad I' = \frac{1mA}{30} \times 20 = \frac{2}{3}mA$

Let x be the effective resistance of the circuit.

$\therefore \quad 3V = 3000\Omega \times 1mA = x\Omega \times \frac{2}{3}mA$

$\Rightarrow \quad x = 4500\Omega$

$\therefore \quad$ Resistance to be added = (4500 – 50) Ω

= 4450 Ω

≈ 4440 Ω

30. (a)

For a circular current carrying loop of radius R

$B_{centre} = \frac{\mu_0 i}{2R}$

$B_{axis} = \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$ [at a distance x from the centre]

Given $B_{centre} = 5\sqrt{5}\ B_{axis}$

$\frac{\mu_0 i}{2R} = 5\sqrt{5}\ \frac{\mu_0 i R^2}{2(R^2 + x^2)^{3/2}}$

$\Rightarrow \quad \frac{(R^2 + x^2)^3}{R^6} = 125$

$\Rightarrow \quad \frac{R^2 + x^2}{R^2} = 5$

$\Rightarrow \quad \frac{x^2}{R^2} = 4$

$\Rightarrow \quad x = 2R$

$\Rightarrow \quad x = 2 \times 10 = 20$ cm = 0.2m

31. (a)

For equilibrium

$F_B = mg$

Bil = mg

$\Rightarrow \quad m = \frac{Bil}{g}$

$\Rightarrow \quad m = \frac{(0.5) \times (2.5) \times (50 \times 10^{-2})}{10} = 62.5g$

Copyright © Think and Learn Pvt. Ltd.

32. (d)
A bar magnet can produce magnetic field. Option (d) is False

33. (c)
Time period of oscillation

$$T = 2\pi\sqrt{\frac{I}{MB}}$$

$$= 2\pi\sqrt{\frac{12\times10^{-6}}{(6\times10^{-2})(2\times10^{-2})}}$$

$$= 2\pi\sqrt{\frac{12\times10^{-6}}{12\times10^{-4}}}$$

$$= 2\pi \times 10^{-1}$$

$\therefore$ For 20 oscillations

$T` = 20 \times 2\pi \times 10^{-1}$

$T` = 12$ sec

34. (a)
Susceptibility of Ferromagnetic substance is >>>1, because it is strongly attracted in an external magnetic field

35. (c)
If the magnet is falling vertically down with its north pole towards the coil, then upper face of coil will become north pole, So that it can oppose the approaching north pole of the magnet. Here, the acceleration will be less than "g"

36. (b)
Working of magnetic breaking in train is based on eddy current

37. (a)
Voltage difference $= B\ell V \sin\theta$

$= (4 \times 10^{-4}) \times (20) \times (400) \sin30^{o}$

$= 1.6$ V

38. (b)
When iron rod is inserted in the coil, then the inductive increases, hence the current flowing through the bulb decrease, therefore brightness decreases.

39. (c)
$P = V.I.$

$\Rightarrow \quad I = \frac{P}{V}$

$\Rightarrow \quad I = \frac{12}{48} = 0.25$ A

$\therefore$ Peak current $= \sqrt{2}\ I$

$$= \sqrt{2}\times\frac{1}{4}$$

$$= \frac{1}{2\sqrt{2}}\text{ A}$$

40. (b)

Copyright © Think and Learn Pvt. Ltd.

$\omega = 2\pi f$

$\omega = 2 \times 3.14 \times 50$

$\omega = 314$

$\tan\phi = \frac{xL}{R} = \frac{\omega L}{R} = \frac{1/\sqrt{3}}{1} = \frac{1}{\sqrt{3}}$

$\Rightarrow \phi = 30^{o}$

$\Rightarrow \quad \omega t = \frac{\pi}{6}$

$\Rightarrow \quad t = \frac{\pi}{6 \times \omega}$

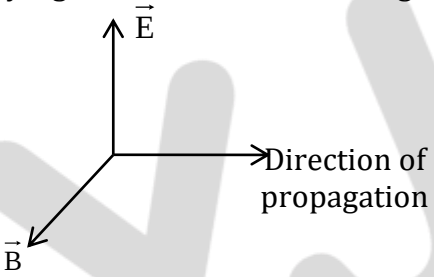
$\Rightarrow \quad t = \frac{3.14}{6 \times 314}$

$t = \frac{1}{600}$ sec

41. (c)

As we know, by property of electro-magnetic wave, the electric field, magnetic field & the direction of propagation, all the three are mutually perpendicular.

There by right hand thumb rule using the figure. $\vec{E} \times \vec{B}$



42. (c)

In ray optics, all the directions in the direction of incident ray are taken as positive

43. (b)

Using mirror formula

$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$\Rightarrow \quad f = \frac{R}{2} = \frac{15}{2}$

$\Rightarrow \quad \frac{1}{-15/2} = \frac{1}{-10} + \frac{1}{v}$

$\Rightarrow \quad V = -30\text{cm}$

$\Rightarrow \quad h_i = -3 \times 10$

$\Rightarrow \quad h_i = -30$

$\therefore$ Image will be magnified & inverted

44. (c)

During scattering of light

Intensity: $I \propto \frac{1}{\lambda^4}$

45. (b)

Copyright © Think and Learn Pvt. Ltd.

Fringe width: $w = \frac{D\lambda}{d}$

$\Rightarrow \quad w \propto \lambda$

But $\lambda_{\text{yellow light}} > \lambda_{\text{Blue light}}$

$\Rightarrow$ Interference pattern will become narrower

46. (a)

Refractive index of medium

$\mu = \frac{c}{v}$

$C \rightarrow$ speed of light in vacuum or air

$V \rightarrow$ speed of light in medium

Above equation implies that speed of light decrease when it goes from rarer medium (air) to denser medium

Also; $\lambda = \frac{v}{f}$

λ = wave length, $f \rightarrow$ frequency

Frequency remains the same for all medium unless source of light is disturbed

$\Rightarrow \quad \lambda \propto V$

$\Rightarrow$ Wave length decreases.

47. (a)

From, Malus law

$I = I_0 \cos^2 \theta$

There; $I = \frac{I_0}{2}$

$\Rightarrow \quad \frac{I_0}{2} = I_0 \cos^2\theta$

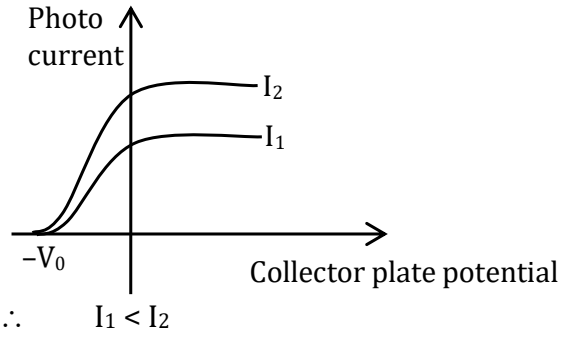
$\text{Cos}\theta = \frac{1}{\sqrt{2}}$

$\Rightarrow \quad \theta = 45^o$

48. (c)

The value of photo current depends on intensity of incident light. So, when the intensity of incident light increases, then the value of photo current is increases.

Intensity of light α photo current



$\therefore \quad I_1 < I_2$

49. (d)

Copyright © Think and Learn Pvt. Ltd.

From de-broglie expression

$\lambda = h/mv$

$\lambda = \dfrac{h}{m\sqrt{2gH}}$

$\Rightarrow \quad \lambda \propto H^{-1/2}$

50. (b)

Rutherford is credited with the discovery of nucleus in an atom.

51. (a)

The expression of energy is

$E_n = \dfrac{13.6}{n^2}$

$\therefore$ required energy

$E = E_2 - E_4$

$= \dfrac{13.6}{(2)^2} - \dfrac{13.6}{(4)^2}$

$= +2.55$

52. (b)

In a nuclear reactor, the function of moderator is to decrease the speed of neutrons.

53. (a)

$^{238}_{92}U \xrightarrow{\alpha} {}^{234}_{90}U \xrightarrow{\beta} {}^{234}_{91}U \xrightarrow{\beta} {}^{234}_{92}U$

$\therefore$ 1 α and 2β particles

54. (b)

Mass defect of ^4_2He ; $\Delta M = 0.03u$

$\therefore$ Binding energy of helium: $E = \Delta M \times 931$ MeV

$E = 0.03 \times 931.5 = 27.93$ MeV

$\therefore$ Number of nucleons in helium; $n = 4$

Binding energy per nucleon = $\dfrac{E}{n} = \dfrac{27.93}{4} = 6.9825$

55. (d)

In Germanium energy gap is less than 3eV

56. (c)

Zener diode is used as a voltage regulated as in reverse biased condition it exhibits constant voltage

57. (a)

In case of transistor, emitter is of moderate size and heavily doped so as to supply large number of majority charge carriers for current flow.

58. (b)

Copyright © Think and Learn Pvt. Ltd.

The voltage drop across AB by using KVL is as follows:

$V_{AB} = (0.2\times10^{-3}\times5\times10^{3})+0.2+(0.2\times10^{-3}\times5\times10^{3})$

= 1 + 0.2 + 1

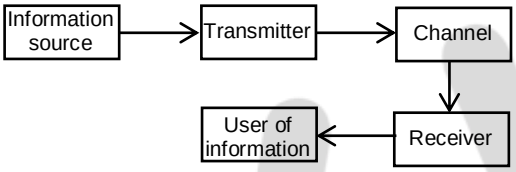
= 2.2V

59. (d)

NAND and NOR gate are considered as universal gates. Out of the given options, an option (D) is NAND gate.

60. (b)

The block diagram of communication system is as follows:



Copyright © Think and Learn Pvt. Ltd.